Venzini

Project Proposal

An AI-Assisted Yield Management Platform for Data-Driven Optimization in the Petroleum Industry

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1. Project overview

Venzini is an AI-assisted web platform for petroleum distributors. It forecasts fuel demand across a network of stations and helps plan deliveries: how many litres to deliver, which stations to prioritize, and how to route the available tankers.

It addresses three common problems: stations running out of fuel, excess inventory, and unnecessary or costly delivery trips. Every recommendation is explained and must be approved by a manager.

2. Objectives

- Reduce stockouts at stations.
- Reduce excess inventory.
- Reduce unnecessary delivery trips and route fuel costs.
- Support managers with clear, explainable recommendations they stay in control of.

3. In scope

The platform is built around the following modules:

- **Network overview:** map of all stations with tank levels and stockout-risk alerts.
- **Demand forecasting:** 7-day forecast per station and fuel type, based on sales history, previous orders, tank levels, weather, local events and seasonality, with confidence level and influencing factors; plus daily totals for the whole network and for each warehouse.
- **Urgency and priority:** station risk (high, medium, low) and a delivery priority ranking based on risk and available resources.
- **Resource management:** the dispatcher uploads the tankers (with their compartments), drivers and warehouse fuel stock available each day, and records the actual departure and return time of each trip.
- **Delivery planning:** recommended quantities and optimal multi-tanker routes, one trip per tanker, with each station served from its closest warehouse.
- **Route cost:** fuel cost of each tanker route, in US dollars, included in the optimization.
- **Route map and editing:** interactive map where the manager can reorder stops and move stations between tankers.
- **Risk analysis:** warnings with severity levels whenever the manager changes the optimal plan; an overloaded tanker blocks approval until it is fixed.
- **Recommendations and approval:** review, adjust, approve or reject each action or the whole plan; rejecting a station triggers a new optimal plan.
- **Decision history:** a read-only record of every decision, visible to all roles.
- **Administration:** user accounts and roles (system admin); station records (system admin and operations manager).

3.1 MVP

The MVP is a clickable frontend prototype that uses fake data. It shows every main screen and flow so the idea can be tested and demonstrated before the logic is built:

- Login with three roles: operations manager, dispatcher and system admin, each seeing only its own screens.
- Network overview, station forecast and forecast totals.
- Resources and trips pages for the dispatcher.
- Delivery planner with the route map, drag and drop, risk warnings and costs.
- Review and approve page, and decision history.

- Administration pages for users and stations.

No real calculations or saved data are required in the MVP.

3.2 Version 1 core (after the MVP)

The logic behind the MVP screens:

- Simulated 7-day forecasts with confidence level, influencing factors and totals.
- Station risk and delivery priority, using a simple priority rule.
- Recommended quantities and multi-tanker routes that respect tank, compartment, tanker and warehouse limits and delivery windows.
- Route cost and live recalculation after each edit, with risk analysis.
- Approval per action or for the whole plan, re-planning after a rejection, and the decision log.
- Resource upload, trip time recording and role-based access.

3.3 Rest of version 1

- Editing delivery quantities, with the allowed maximum shown and saving blocked above it.
- Automatic new optimal plan when quantities are edited.
- Comparison of estimated and actual trip times.
- Full yield-management priority logic (version 1 core uses a simpler priority rule).
- Station management by the system admin and the operations manager (version 1 core uses preloaded stations).

4. Out of scope

- Setting pump prices.
- Controlling fuel systems or equipment.
- Access for tanker drivers (in any version).
- Access for data providers (a dedicated role is planned for a later version).
- Real distributor data and live integrations (version 1 uses sample data and simulations).
- Live tanker tracking (actual trip times are entered by the dispatcher).
- Large language models (LLMs) and autonomous AI agents.
- Handing the approved plan to drivers and the depot inside the platform.
- Mobile application (version 1 targets desktop browsers).

5. Users and roles

- **Operations manager:** Views the network, forecasts and totals, plans deliveries, edits routes and quantities, manages station records, and approves or rejects recommendations. Cannot change the resources uploaded by the dispatcher.
- **Dispatcher:** Uploads the resources available for each delivery day (tankers, drivers, warehouse fuel stock), records actual trip times, and reads the forecast totals per warehouse. Has no access to plans or station forecasts.
- **System admin:** Manages user accounts, roles and station records.

All three roles can view the decision history, read-only.

6. Deliverables

- Project scope (this document)
- Requirements elicitation document
- SRS (Software Requirements Specification)

- SDD (Software Design Document)
- MVP (clickable prototype with fake data)

7. Assumptions

- Sample data is realistic enough to test and demonstrate the platform.
- Every recommendation is reviewed and approved by a manager before it is acted on.
- The dispatcher uploads resources before planning starts.
- Urgency is split into station risk (based on stock and forecast) and delivery priority (combining risk with resources and distance).
- The fuel price is public, and all costs are in US dollars.

8. Constraints

- Solo developer.
- No access to real distributor data; forecasts, routes and costs are simulated.

9. Future versions

- Dedicated interface for station managers.
- Integration with real data sources.
- Data provider and data manager role to upload external and operational data.
- Use of actual trip and unloading times to improve time estimates.

10. Success criteria

- MVP: every main screen and flow can be clicked through with fake data, for each role.
- All version 1 core features work end to end with sample data.
- Every forecast, recommendation and risk is explained to the user.
- Route edits update live and show their risks and cost impact.
- No action is taken without manager approval, and every decision is logged.

Detailed, measurable criteria are defined in the SRS (version 1.0).